

# Brain Tumor Segmentation

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## 1. Problem Overview

The objective of this project was to automatically segment brain tumors from MRI scans using deep learning. Unlike classification, which only predicts whether a tumor is present, segmentation identifies the **exact pixels corresponding to the tumor region**. This is crucial for medical analysis tasks such as tumor size estimation, treatment planning, and disease monitoring. The BraTS 2021 dataset was used, which provides multi-modal MRI scans along with expert-annotated segmentation masks.

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## 2. Dataset Description

The BraTS 2021 dataset consists of 3D brain MRI volumes stored in NIfTI (.nii.gz) format. Each patient scan contains multiple MRI modalities and a corresponding ground-truth segmentation mask. In this project, only the **FLAIR modality** was used as input because it highlights tumor regions clearly, making it suitable for segmentation tasks. The segmentation mask contains labeled tumor regions, which were simplified into a binary format distinguishing tumor and non-tumor areas.

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## 3. Data Preprocessing

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Since MRI scans are three-dimensional volumes, direct training on 3D data would be computationally expensive. Therefore, each 3D MRI volume was converted into multiple **2D axial slices**, treating each slice as an independent training sample. This significantly increased the number of training examples while keeping the model lightweight.

Each slice was resized to a fixed resolution and normalized so that pixel intensity values lay between 0 and 1. The corresponding segmentation masks were also resized and converted into binary masks, where tumor pixels were assigned a value of 1 and background pixels were assigned 0. Slices without any tumor were discarded to prevent the model from being biased toward background predictions. After preprocessing, a total of **620 valid MRI slices** were obtained for training.

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## 4. Model Architecture (U-Net)

A **U-Net architecture** was implemented using PyTorch for the segmentation task. U-Net is a convolutional neural network designed specifically for biomedical image segmentation. It consists of an **encoder path** that extracts high-level features by progressively downsampling the image and a **decoder path** that reconstructs the segmentation mask by upsampling feature maps.

Skip connections between corresponding encoder and decoder layers were used to preserve spatial information lost during downsampling. This allows the model to learn both **what** the tumor looks like and **where** it is located in the image. The final layer outputs a probability map indicating the likelihood of each pixel belonging to the tumor class.

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## 5. Model Training

The model was trained using **Binary Cross-Entropy loss**, which is suitable for binary segmentation tasks. The Adam optimizer was used due to its stability and fast convergence.

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Training was performed for five epochs using GPU acceleration. During training, the loss value decreased consistently from approximately **0.31 to 0.04**, indicating that the model successfully learned meaningful tumor representations from the data.

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## 6. Evaluation Metrics

To evaluate segmentation performance, two standard metrics were used: **Dice Score** and **Intersection over Union (IoU)**. These metrics measure the overlap between the predicted segmentation mask and the ground-truth mask and are more informative than accuracy for segmentation tasks with class imbalance.

The trained model achieved a **Dice Score of approximately 0.77** and an **IoU score of approximately 0.62**, indicating strong agreement between predicted tumor regions and ground-truth annotations.

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## 7. Visualization of Results

Qualitative evaluation was performed by visualizing the original MRI slice, the corresponding ground-truth segmentation mask, and the predicted segmentation mask side by side. The visual results show that the model successfully identifies tumor regions and captures their spatial structure, further validating the quantitative performance metrics.

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## 8. Conclusion

In this project, a deep learning-based brain tumor segmentation system was successfully developed using the BraTS 2021 dataset. Through careful preprocessing, a U-Net

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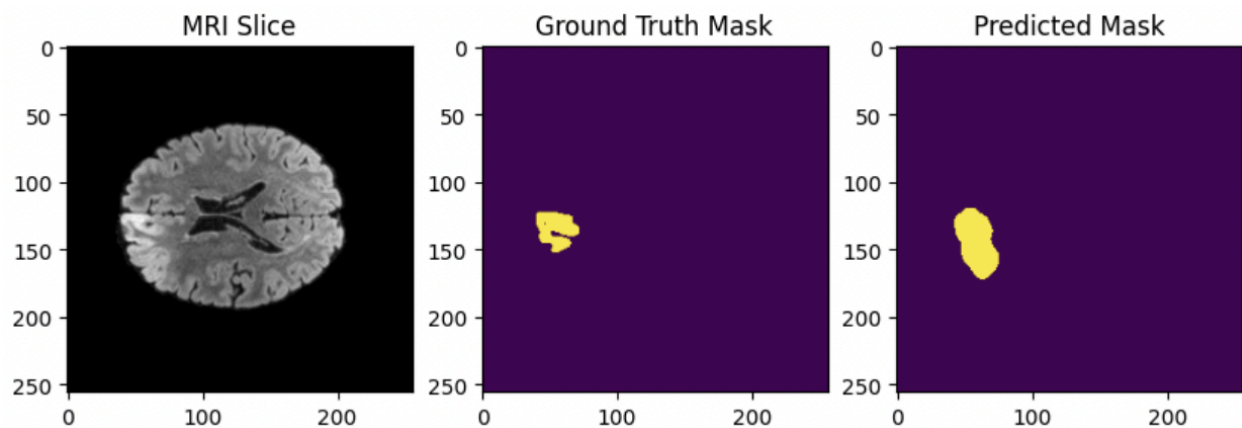
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architecture, and appropriate evaluation metrics, the model demonstrated effective tumor segmentation performance. This work highlights the potential of deep learning techniques in assisting medical image analysis and automated diagnostic workflows.

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```
Epoch 1/5 | Loss: 0.3149  
Epoch 2/5 | Loss: 0.0566  
Epoch 3/5 | Loss: 0.0488  
Epoch 4/5 | Loss: 0.0458  
Epoch 5/5 | Loss: 0.0441
```

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... Dice Score: 0.7654061913490295  
IoU Score : 0.6199660301208496
```



## 9. Imp Links

Report :

<https://docs.google.com/document/d/1aq3iFLtojDkjNvHdH6MsW8Tkk4c9L6j3SF5vvB6nW1o/edit?usp=sharing>

Google colab code :

[https://colab.research.google.com/drive/1uyZqzLrqWjHWj1dwH2Jltla6NWeRV\\_vw?usp=sharing](https://colab.research.google.com/drive/1uyZqzLrqWjHWj1dwH2Jltla6NWeRV_vw?usp=sharing)